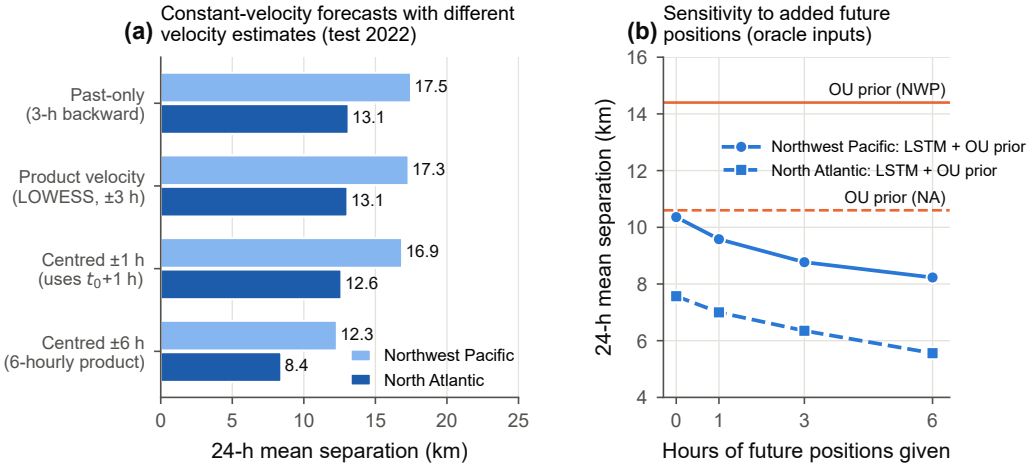
(a)
Constant-velocity forecasts with different velocity estimates (test 2022)
Past-only (3-h backward)
17.5
13.1
Product velocity (LOWESS, ±3 h)
17.3
13.1
Centred ±1 h (uses t_0+1 h)
16.9
12.6
Centred ±6 h (6-hourly product)
12.3
8.4
Northwest Pacific
North Atlantic
0
5
10
15
20
25
24-h mean separation (km)
(b)
Sensitivity to added future positions (oracle inputs)
OU prior (NWP)
Northwest Pacific: LSTM + OU prior
North Atlantic: LSTM + OU prior
OU prior (NA)
24-h mean separation (km)
16
14
12
10
8
6
4
0
1
3
6
Hours of future positions given